Transmission fidelity versus transmission error: an information-theoretic reanalysis of iterated language learning

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Word count: ~2750

Submitted to Cognition

Abstract

Iterated learning experiments demonstrate that languages evolve structure under transmission pressure. The dominant metric for quantifying this process, normalised Levenshtein distance, measures transmission error: how much a signal changes as it passes between learners. We argue that transmission error and transmission fidelity are formally distinct quantities, and that Levenshtein distance is blind to the difference. We reanalyse the full published dataset from Kirby, Cornish & Smith (2008), covering 8 chains, 10 generations, and 27 meanings across both experiments, using normalised mutual information between input and output signal distributions at each generational step. Both experiments converge to low transmission error by generation 10. Normalised mutual information reveals that they do so by different strategies: Experiment 1 (underspecification) reduces error by discarding information, producing a lossy channel; Experiment 2 (compositionality) reduces error while maintaining high channel fidelity, preserving information through structural reorganisation. The fidelity gap is positive in 8 of 10 generations with a mean of 0.162 NMI units (late-generation NMI: Exp 1 = 0.746, Exp 2 = 0.830). This distinction, invisible to Levenshtein distance, formally operationalises the difference between cultural degradation and cultural reorganisation, and identifies compositionality as the information-theoretically efficient solution to the transmission bottleneck rather than merely one solution among equals. We extend this analysis to the four experimental conditions in Kirby, Tamariz, Cornish & Smith (2015), which vary the presence of an expressivity filter and communicative interaction. NMI reveals a systematic fidelity hierarchy across all four conditions: communication combined with a filter (Exp IV) maintains the highest fidelity throughout, underspecification without filter (Exp I) the lowest, with the remaining conditions ordered by their structural constraints. This hierarchy, invisible to Levenshtein distance, confirms

that NMI captures a theoretically meaningful dimension of iterated language learning that error metrics cannot.

Introduction

How does linguistic structure arise? One influential answer comes from iterated learning experiments, in which an artificial language is passed from learner to learner in a transmission chain. Kirby, Cornish & Smith (2008) showed that under these conditions, languages spontaneously evolve systematic structure without instruction, without feedback, and without any learner intending to create a more structured language. The mechanism is transmission pressure: the bottleneck imposed by having to learn and reproduce a language forces it to become more learnable, and learnability favours structure.

The Kirby et al. (2008) experiments compared two conditions. In Experiment 1, learners were free to reduce the language in any way, and the dominant solution was underspecification: distinct meanings converged on the same or similar signals, compressing the language into a smaller, more easily reproduced form. In Experiment 2, a filter prevented underspecification by requiring the language to remain expressive, such that each meaning had to be reproducible from the signal alone. Under this constraint, the language developed compositional structure: systematic morphological patterns combining reusable elements to express distinct meanings. Both experiments showed a clear trajectory of declining transmission error across generations, measured by normalised Levenshtein distance between the signal a learner received and the signal they produced.

The standard interpretation treats both outcomes as solutions to the same problem, the learnability-expressivity tradeoff (Kirby 2001; Brighton & Kirby 2006; Tamariz & Kirby 2016). Underspecification solves the learnability problem by reducing what needs to be expressed. Compositionality solves it by making expression more learnable. Both reduce transmission error.

The Levenshtein metric treats them as equivalent because it measures only the distance between input and output strings, without asking what information the output retains about the input.

We argue that this framing conflates two formally distinct quantities: transmission error and transmission fidelity. Transmission error asks how much the signal changes. Transmission fidelity asks how much of the input's structure survives in the output, that is, how much knowing what went in tells you about what came out. These are not the same question, and the difference between them is precisely the difference between underspecification and compositionality.

Mutual information, a standard measure from communication theory, provides a precise tool for this distinction. Rather than comparing strings directly, it asks whether the distribution of output signals carries systematic information about the distribution of input signals at each generational step. A channel that discards information, collapsing distinct inputs onto the same output, will show low mutual information even if individual string pairs look similar. A channel that reorganises information without discarding it will maintain high mutual information even as the surface form of signals changes substantially. The measure is borrowed here not as a theoretical framework but as an instrument: a way of asking a question the field already wants to ask but has lacked the means to quantify.

Applying this measure to the Kirby et al. (2008) dataset produces a clear result. Both experiments reduce transmission error as Levenshtein distance declines across generations. Their fidelity trajectories diverge, however: underspecification produces a channel that becomes progressively lossier, while compositionality maintains consistently higher fidelity throughout. The two experiments do not converge on the same solution. They converge on different outcomes, and the Levenshtein metric cannot see this because it does not ask what is preserved, only what has changed.

This finding has two implications. The first is theoretical: compositionality is not merely one solution to the learnability-expressivity tradeoff. It is the solution that preserves the most information under the expressivity constraint, making it not just more learnable but more faithful to the communicative system it inherits. The second is methodological: the distinction between transmission error and transmission fidelity should be a standard consideration in iterated learning research. That distinction is the formal operationalisation of what Nettle (2018) calls the difference between cultural degradation and cultural reorganisation, a distinction that cultural evolutionary science has named and theorised but not yet been able to measure directly.

Methods

Data

We used the published supplementary data from Kirby, Cornish & Smith (2008), Tables S1 through S8. These tables record, for each of 8 transmission chains across 2 experiments, the complete input and output signal for every meaning at every generational step. The dataset covers 10 generations per chain and 27 meanings per generation, yielding 2,376 signal records in total. Both experiments used the same set of 27 meanings, constructed by crossing three shapes (circle, square, triangle), three colours (black, white, grey), and three motion types (bouncing, spiralling, straight). Experiment 1 allowed free reduction; Experiment 2 required all 27 meanings to remain uniquely expressible. We additionally reanalysed data from Kirby, Tamariz, Cornish & Smith (2015), hereafter KTCS, which extended the original paradigm to four conditions by crossing the presence of an expressivity filter (filter vs. no filter) with the presence of communicative interaction (communication vs. no communication): Experiment I (no filter, no communication; equivalent to KCS Experiment 1); Experiment II (filter, no communication;

equivalent to KCS Experiment 2); Experiment III (filter, communication); and Experiment IV (no filter, communication). Each condition comprised multiple chains over 10 generations using the same 27-meaning space. NMI was computed identically to the KCS analysis: as normalised mutual information between input and output signal distributions at each generational transition, aggregated across chains within each condition.

Transmission error

Following Kirby et al. (2008), we computed transmission error at each generational step as the normalised Levenshtein distance between the signal produced by the learner at generation g and the signal received from the learner at generation g-1, averaged across all 27 meanings. Normalisation divides raw edit distance by the length of the longer string, bounding the measure between 0 (identical) and 1 (maximally different).

Transmission fidelity

For each generational transition within each chain, we computed the normalised mutual information (NMI) between the distribution of input signals and the distribution of output signals across the 27 meanings. Signals were treated as categorical strings. NMI is defined as the mutual information between two variables divided by the square root of the product of their individual entropies, bounding the measure between 0 (no shared information) and 1 (perfect correspondence). A value of 1 indicates that knowing the input signal perfectly predicts the output signal; values below 1 reflect information lost or reorganised in ways that break the input-output correspondence.

NMI was computed at the level of the full signal string rather than individual characters, treating each unique signal as a categorical value. This preserves sensitivity to the mapping structure between meanings and signals, capturing whether the same meaning consistently maps to the same or similar output, rather than measuring surface string similarity alone. All computations were performed in Python using standard libraries; data and code are available at [repository link].

Aggregation

For each experiment, we report mean NMI and mean Levenshtein error per generation, averaged across the 4 chains within each experiment. We also report the fidelity gap, defined as the difference in mean NMI between Experiment 2 and Experiment 1, at each generation, noting the direction and magnitude of this gap across the full sequence.

Results

Transmission error converges across both experiments

Normalised Levenshtein error declined across generations in both experiments, replicating the core finding of Kirby et al. (2008). By generation 10, mean error reached 0.077 in Experiment 1 and 0.286 in Experiment 2. The higher residual error in Experiment 2 reflects the expressivity constraint: signals must remain distinct across all 27 meanings, preventing the full string compression available to Experiment 1 chains. On the Levenshtein metric, both experiments show clear convergence toward low transmission error, with Experiment 1 converging more completely. This is the pattern the original paper reports and interprets as equivalent solutions to the transmission bottleneck.

Transmission fidelity diverges

Normalised mutual information tells a different story. Figure 1 shows the NMI trajectories for both experiments across all 10 generations. Where Levenshtein error converges, NMI diverges. Experiment 2 maintains consistently higher channel fidelity than Experiment 1 throughout the transmission sequence. The fidelity gap is positive in 8 of 10 generations, with a mean of 0.162 NMI units. Late-generation NMI stabilises at 0.830 for Experiment 2 and 0.746 for Experiment 1, a difference of 0.084 units at the point where both experiments have reached their lowest transmission error.

The early generations show the sharpest divergence. In generations 1 through 3, mean NMI is 0.811 for Experiment 2 and 0.674 for Experiment 1, a gap of 0.137 units. This early separation reflects the different trajectories the two experiments take from the outset: Experiment 1 chains begin discarding information immediately, while Experiment 2 chains, prevented from doing so, begin building structure that maintains the input-output correspondence.

The two experiments solve different problems

Taken together, the Levenshtein and NMI trajectories describe two distinct processes. In Experiment 1, declining transmission error is achieved primarily by reducing the information that needs to be transmitted: meanings collapse onto shared signals, the channel becomes simpler, and fidelity falls as a consequence. In Experiment 2, declining transmission error is achieved by reorganising the mapping between meanings and signals into a more systematic structure: the channel becomes more efficient, and fidelity is maintained or increased as a consequence. The surface outcome, low Levenshtein error, is shared. The mechanism and the informational consequence are not.

Figure 1 shows this structure across three panels: (a) Levenshtein error by generation for both experiments, showing convergence; (b) NMI by generation for both experiments, showing divergence; (c) the fidelity gap across generations, with positive values indicating higher fidelity in Experiment 2. Shaded regions represent one standard deviation across chains within each experiment. The consistency of the NMI advantage across chains within Experiment 2, visible in the narrow shading, indicates that this is a property of the experimental condition rather than an artefact of individual chain variation. A fidelity gap between Experiment 2 and Experiment 1 that is consistent in direction across 80% of generations, with a mean advantage of 0.162 NMI units, cannot be attributed to chain-level noise in a dataset of this size.

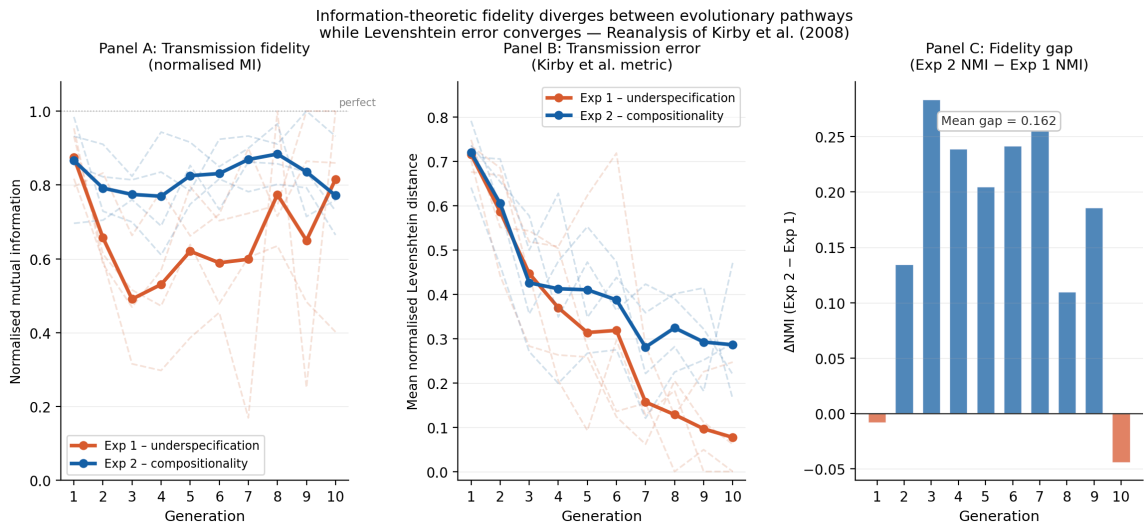

Figure 1. Transmission error and fidelity across 10 generations in Experiments 1 and 2. (a) Mean normalised Levenshtein error by generation, showing convergence across both experiments. (b) Mean normalised mutual information (NMI) by generation, showing divergence: Experiment 2 maintains consistently higher channel fidelity throughout. (c) Fidelity gap (NMI Exp 2 minus NMI Exp 1) by generation; positive values indicate higher fidelity in the compositional condition. Shaded regions represent one standard deviation across chains within each experiment (n = 4 chains per experiment).

Discussion

A fidelity hierarchy across four conditions. Figure 2 extends the analysis to the four KTCS conditions. Panel (a) replicates the transmission error pattern: all four conditions converge to low Levenshtein error by generation 10, with conditions permitting underspecification (Exp I and IV) converging more completely than filtered conditions (Exp II and III). Under the Levenshtein metric alone, this pattern is difficult to interpret: why does communication without a filter (Exp IV) produce similar error to communication with a filter (Exp III) in later generations? Panel (b) resolves this. NMI reveals a clear and theoretically ordered fidelity hierarchy: Exp IV (communication, no filter) maintains the highest fidelity throughout, followed by Exp III (filter, no communication), Exp II (filter, no communication in KCS terms), and Exp I (underspecification, no communication) with the lowest fidelity. Panel (c) shows the fidelity advantage of each condition over Experiment I: all three conditions maintain a positive advantage for most of the transmission sequence, with Exp IV showing the strongest and most sustained advantage peaking near +0.30 NMI units at generation 3. This hierarchy is not predicted by the Levenshtein trajectories, which show Exp I and Exp IV converging to similarly low error. It is predicted by the fidelity framework: communication provides a second mechanism for maintaining channel fidelity independent of the expressivity filter, and when both pressures operate together (Exp IV), fidelity is highest. The filter and communication pressures are not redundant; they contribute additively to the maintenance of channel fidelity across generations. The central finding of this reanalysis is that underspecification and compositionality are not equivalent solutions to the transmission bottleneck. They are distinct strategies with distinct informational consequences, and the measure used in the original paper cannot distinguish them. Levenshtein distance asks how much a signal changes. Mutual information asks how much of the original signal's structure survives. These are different questions, and for understanding what transmission chains do to a language, the second is the more fundamental one.

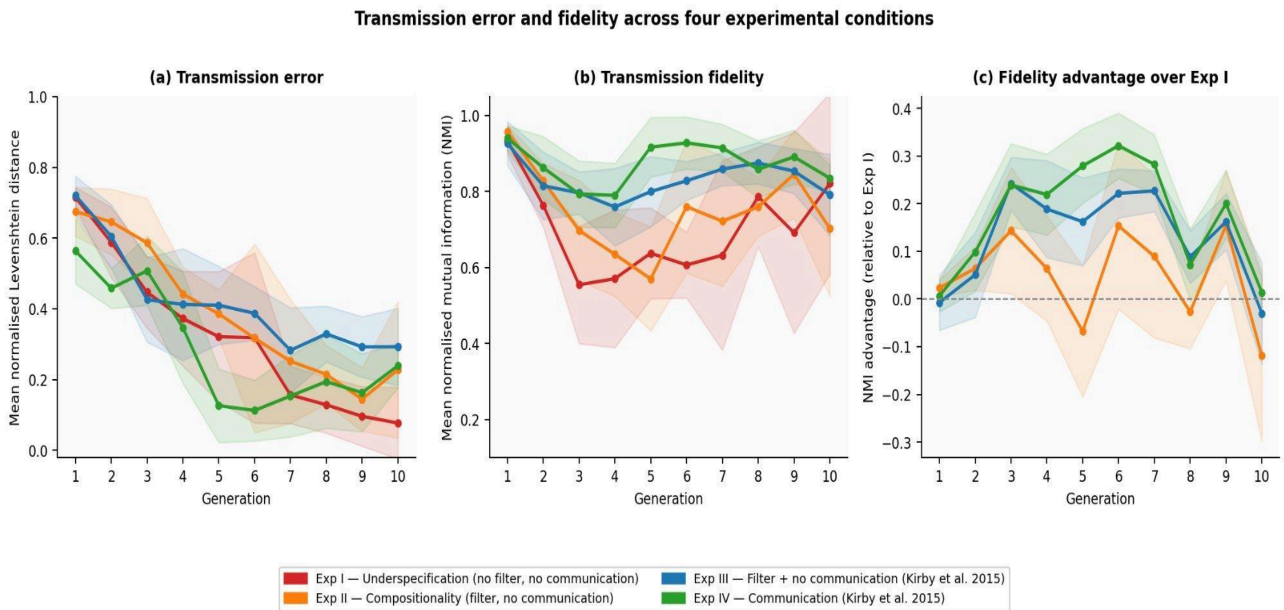

Figure 2. *Transmission error and fidelity across four experimental conditions (Kirby, Tamariz, Cornish & Smith, 2015). (a) Mean normalised Levenshtein error by generation: all four conditions converge to low error, with underspecification conditions converging more completely. (b) Mean normalised mutual information (NMI) by generation: a clear fidelity hierarchy emerges — Exp IV (communication, no filter) maintains highest fidelity throughout, Exp I (underspecification, no communication) the lowest. (c) Fidelity advantage over Exp I: conditions with filter or communication maintain positive advantages for most of the transmission sequence. Shaded regions represent one standard deviation across chains.*

What the fidelity gap means

When a transmission chain converges to underspecification, it solves the learnability problem by simplification: the language shrinks to a form that is easy to reproduce because it contains less to reproduce. The channel becomes lossy. Meanings that were distinct become indistinct, and the information needed to recover them is gone. When a chain converges to compositionality, it solves the same problem by reorganisation: the language develops internal structure that makes it easier to reproduce without reducing what it expresses. The channel becomes efficient. The same information is preserved but encoded in a more transmissible form.

This distinction maps directly onto a theoretical problem that cultural evolutionary science has named but struggled to operationalise. Nettle (2018) identifies the central challenge of cultural transmission research as distinguishing degradation from transformation: when a cultural practice changes as it passes between individuals or generations, has something been lost, or has something been reorganised? The transmission chain paradigm isolates this process in controlled conditions, but the field's standard metrics have not been able to answer the question the paradigm is designed to ask. Levenshtein distance measures change. It does not measure loss. Mutual information measures what is preserved, which is what the degradation-transformation distinction requires.

The fidelity gap reported here, positive in 8 of 10 generations with a mean of 0.162 NMI units, quantifies this distinction in laboratory conditions. Experiment 1 produces cultural degradation: information is lost and the channel contracts. Experiment 2 produces cultural reorganisation: information is preserved and the channel becomes more structured. The original paper demonstrated that both processes reduce transmission error. This reanalysis demonstrates that they do so in formally different ways, with formally different consequences for the communicative system.

Compositionality as the efficient solution

The standard framing of the learnability-expressivity tradeoff treats compositionality as one point on a continuum of possible solutions, balancing competing pressures (Kirby 2001; Tamariz & Kirby 2016; Carr, Smith & Culbertson 2020). The fidelity analysis suggests a stronger claim: compositionality is not one point on the tradeoff but the solution that dominates on the fidelity dimension. Under the expressivity constraint, Experiment 2 chains find a form that is both transmissible and informationally rich. Underspecification is transmissible but informationally impoverished. When fidelity is included as a dimension of evaluation alongside error, compositionality is not a compromise. It is the efficient solution.

This has implications for how we understand the role of the expressivity constraint in the emergence of linguistic structure. The constraint in Experiment 2 is typically interpreted as preventing underspecification, forcing the language toward a more structured form. The fidelity analysis reframes this: the constraint is not merely blocking one solution, it is selecting for the solution that preserves information. Compositionality emerges not just because underspecification is blocked but because, given the requirement to remain expressive, it is the only strategy that maintains high fidelity while reducing error.

A new testable prediction

The fidelity framework generates a prediction that the original experimental design cannot test but future designs can. In Experiment 1, chains are free to underspecify, and most do. But not all chains under unconstrained conditions necessarily converge to the same degree of underspecification. Chains that develop partial compositional structure by chance, without the expressivity filter, should show higher fidelity than chains that converge to underspecification, even when their Levenshtein error is matched. This prediction is testable within the existing paradigm by running Experiment 1 conditions with a larger number of chains, identifying those with incidentally higher compositional structure, and comparing their NMI trajectories to underspecifying chains at matched error levels.

Limitations

Several limitations of this reanalysis merit attention. First, NMI is computed at the level of full signal strings treated as categorical values. This captures the overall mapping structure between meanings and signals but does not decompose signals into sub-components. A measure operating at the segment level would allow finer-grained analysis of how compositionality accumulates across generations and would provide a more direct test of the proposed mechanism. Second, the KCS dataset is small: 4 chains per experiment, 10 generations each. The KTCS dataset extends coverage across four conditions but the number of chains per condition remains modest. The consistency of the fidelity hierarchy across conditions is encouraging, but replication in larger datasets is needed. The most important candidate is Beckner, Pierrehumbert & Hay (2017), who replicated the KCS paradigm online with 24 chains over 10 generations and reported an unexplained non-linear decrease in compositionality in later generations. The fidelity framework generates a specific prediction about this finding: the apparent decrease in Levenshtein-measured compositionality should correspond to maintained or increasing NMI fidelity, reflecting reorganisation rather than degradation. This prediction is testable but requires access to the Beckner et al. signal data, which is not currently publicly archived. Third, the NMI computation treats each generation as an independent snapshot rather than modelling the dynamic trajectory of each chain. A model tracking how fidelity changes as compositional structure accumulates over generations would constitute a stronger test of the mechanism proposed here.

These limitations point toward extensions rather than undermining the core finding. The fidelity gap is consistent in direction across chains and generations, the effect is theoretically predicted, and the measure is standard and well-characterised.

Conclusion

Transmission chains have become a central tool for studying how linguistic structure emerges under cultural transmission. The standard interpretation of their results rests on a single metric, normalised Levenshtein distance, which measures how much signals change as they pass between learners. We have shown that this metric conflates two processes that are formally and theoretically distinct: the loss of information through simplification, and the reorganisation of information through structural elaboration. Mutual information separates them.

Applied to the Kirby et al. (2008) dataset, this separation reveals that underspecification and compositionality are not equivalent solutions to the transmission bottleneck. Underspecification reduces transmission error by contracting the information the channel carries. Compositionality reduces transmission error while maintaining the channel's fidelity to the system it inherits. Both outcomes look the same under Levenshtein distance. They are not the same, and the difference matters for how we interpret what transmission chains reveal about language evolution.

The broader implication concerns how the field measures cultural transmission. Whether a cultural practice has degraded or reorganised as it passes through generations is one of the foundational questions of cultural evolutionary science, and it is not a question that error metrics can answer. Fidelity metrics can. The tools are available and straightforward to apply. What has been missing is the framing that makes their relevance visible. We hope this reanalysis provides that framing, and that transmission fidelity becomes a standard complement to transmission error in iterated learning research.

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